

Java Programming Assignment

Data Types, Variables & Operators

Q1

Which of the following is NOT a primitive data type in Java?

- A. int
- B. boolean
- C. String
- D. char

Q2

What will be printed?

```
int x = 7;  
int y = 2;  
System.out.println(x / y);
```

- A. 3 Because this is an integer division, if you want double make at least one var a double
- B. 3.5
- C. 4
- D. Compilation Error

Q3

Which declaration is valid?

- A. float price = 19.99;
- B. float price = 19.99f;
- C. float price = "19.99";
- D. float price = '19.99';

Q4

What is the value of x?

```
int x = 8;  
x %= 3;
```

- A. 0
- B. 1
- C. 2
- D. 3

Q5

Which expression evaluates to false?

- A. `5 == 5`
- B. `10 >= 5`
- C. `8 != 8`
- D. `7 < 9`

Q6

Which operator has the highest precedence?

- A. `+`
- B. `*`
- C. `==`
- D. `&&`

Q7

Which variable name follows Java naming conventions?

- A. `student-name`
- B. `2number`
- C. `studentName`
- D. `class`

Q8

What is the output?

20

```
System.out.println(10 + 5 * 2);
```

Q9

Predict the output.

```
int x = 5;
System.out.println(x++);
System.out.println(x);
```

5
6

Q10

Predict the output.

```
int x = 5;
System.out.println(++x);
System.out.println(x);
```

6
6

Q11

Determine the output.

```
int a = 10;
int b = 3;
System.out.println(a / b);
System.out.println(a % b);
```

3
1

Q12

Predict the output.

true

```
boolean a = true;
boolean b = false;
System.out.println(a || b && false);
```

&& has higher precedence than ||
precedence of logical operators is ! then && then ||

Q13

What will be printed?

13

```
int x = 2, y = 3, z = 4;
System.out.println(x + y * z - 1);
```

Q14

```
int x = 10;
System.out.println(x++ + ++x);
```

22

because $10 + 12 = 22$

Q15

```
int x = 4;
System.out.println(x++ * 2 + --x);
```

12

because $4 * 2 + 4 = 12$

Q16

What is the problem?

```
int age = "20";
```

- A. No error
- B. Wrong data type assigned
- C. Missing semicolon
- D. Invalid variable name

Q17

Identify the error.

```
boolean passed = "true";
```

- A. No error
- B. "true" should be true
- C. Use single quotes
- D. boolean cannot store true/false

Q18

Will this code compile?

```
int x;
x = x + 5;
```

```
System.out.println(x);
```

A. Yes, it prints 5

B. Yes, it prints 0

C. Compilation Error Because x hasn't been initialised with a value

D. Runtime Error

Bonus

Without running the code, determine the output and explain your answer.

```
int a = 5;  
int b = 10;  
System.out.println(a++ + b-- + ++a + --b);
```

This can be interpreted as :

use the current value of a then increment it by 1 later -> a here is 5

+

use the current value of b then decrement it by 1 later -> b here is 10

+

increase first then use the value, a was incremented earlier to 6 so now it becomes 7 -> a here is 7

+

decrement first then use the value, b was decremented earlier to 9 so now it becomes 8 -> b here is 8

so it's $5 + 10 + 7 + 8 = 30$